

Real-time GS Mapping on Aria

Efficient online learning of Gaussian maps

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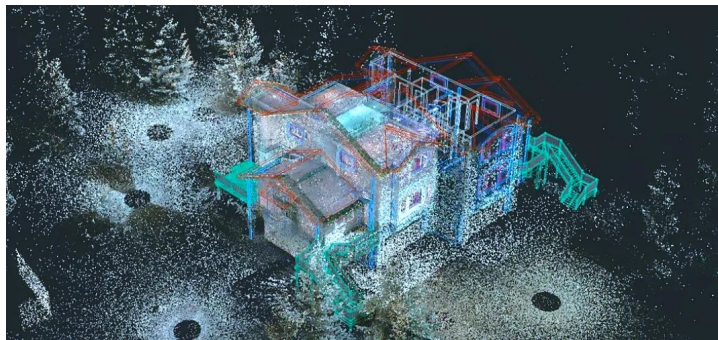
RPM Lab, Seoul National University

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Why Gaussian Splatting?

Build a renderable scene that can be refined with new observations.

Point cloud



Gaussian representation / rendering



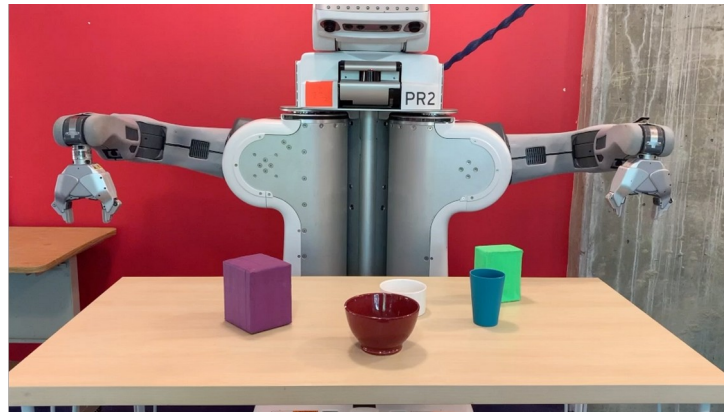
Photorealistic views + an updatable scene representation

Why Real-time Mapping?

A robot or AR user needs a useful map while observations are still arriving.



Observe the environment

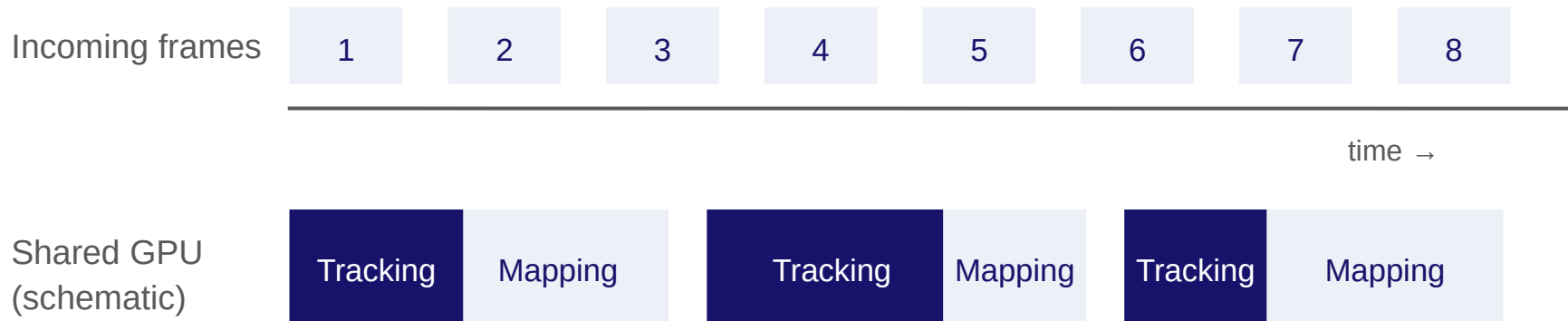


Build and use the map now

Map quality matters when it becomes available.

Mapping under Limited Compute

Online mapping has limited time to learn from growing observations.



How should we spend the available mapping updates?

Use effective observations within available computation.

What Makes Online Mapping Difficult?

New observations compete for updates; appearance alone cannot ensure geometry.

Under-optimized appearance



Too little useful supervision

Incorrect geometry



Opacity in observed free space

How quickly can a newly observed region become reliable?

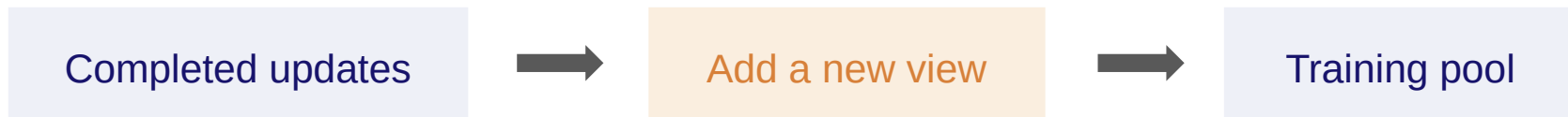
When Should We Add More Views?

More views add information, but leave fewer updates for each view.

Tracking
keyframes



Intermediate RGB frames also carry mapping information.



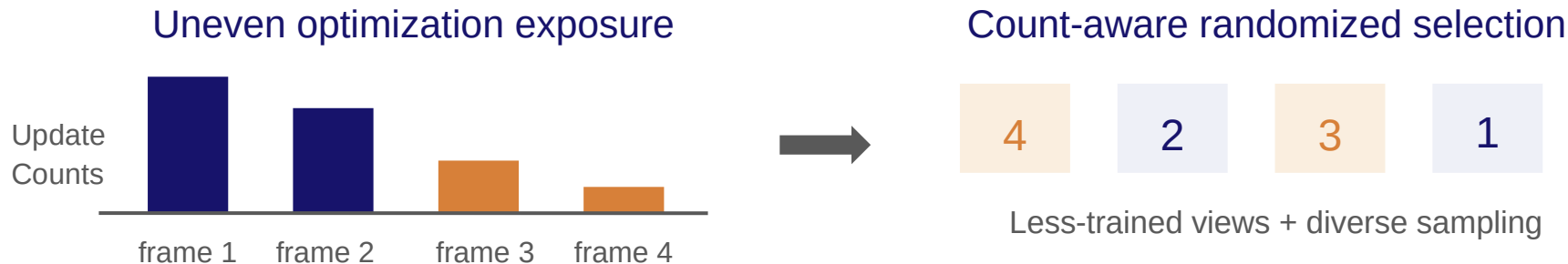
$$N(t) - N_0 \approx \frac{S(t)}{k}$$

N: training-pool size S: completed GPU mapping updates k: updates per added view

Grow the training pool with completed optimization.

Which Views Should We Optimize?

Balance learning opportunities while preserving sampling diversity



Normalized variance
+ entropy

$$\min_{p \in \Delta_N} \mathbb{E}_{i \sim p} [\Delta_i V_{\text{norm}}(\mathbf{n})] - \tau H(p)$$

Softmax solution

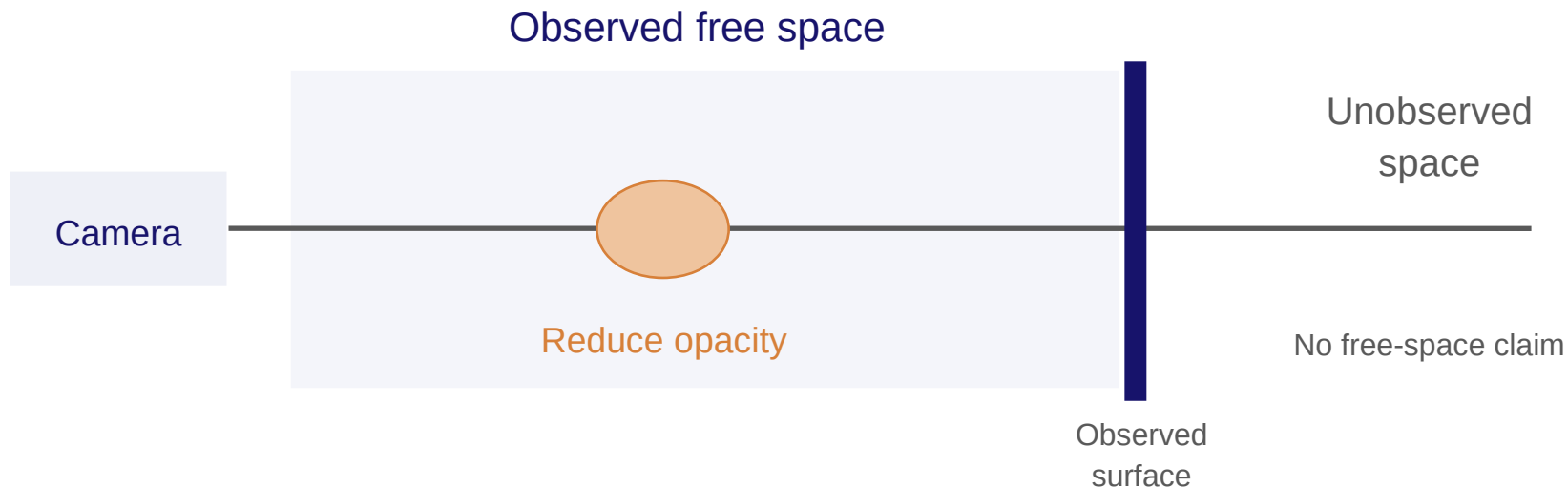
$$p_i^* = \text{softmax}_i(-\beta n_i) = \frac{e^{-\beta n_i}}{\sum_j e^{-\beta n_j}}$$

$$V_{\text{norm}}(\mathbf{n}) = \frac{1}{2N} \sum_i (n_i - \bar{n})^2$$

n_i : selection count H : sampling entropy

Keeping Observed Free Space Empty

RGB alignment can still hide an incorrect 3D structure.



Use causal depth evidence to suppress premature opacity.

Current Progress & Evaluation

View scheduling and geometry are improved.

Baseline



Ours



Quantitative Metrics

Metric	Method	RPNG	UTMM	Aria
PSNR $\uparrow$	VIGS-SLAM	22.49	18.73	22.99
	Ours	24.09	19.25	25.44
SSIM $\uparrow$	VIGS-SLAM	0.7460	0.6441	0.7840
	Ours	0.8052	0.6634	0.8331
LPIPS $\downarrow$	VIGS-SLAM	0.2426	0.4216	0.4555
	Ours	0.1789	0.4123	0.3451

matched-budget appearance + geometry evaluation

Takeaway & Next Step

Build better maps through efficient online optimization.

01

When to add

Pace view growth

02

What to optimize

Balance exposure

03

Where to constrain

Use free-space evidence

Next: validate appearance and geometry gains at the native input rate.